CSE 4632: Digital Signal Processing Lab

Lab 0: Introduction to Python

Objective

- To familiarize students with basic Python programming.
- To understand how Python can be used for signal processing tasks.
- To perform simple numerical operations and visualize data.

Introduction

Python is a powerful and easy-to-learn programming language widely used in engineering and signal processing. In DSP, Python helps us analyze, manipulate, and visualize signals efficiently.

Common libraries:

- **NumPy**: For numerical operations
- **Matplotlib**: For plotting graphs

Getting Started

You can use:

- Jupyter Notebook
- Google Colab
- Any Python IDE (VS Code, PyCharm)

Basic Python Tutorial

1. Variables and Data Types

```
x = 10          # Integer
y = 3.14        # Float
name = "DSP"    # String
```

2. Basic Operations

```
a = 5
b = 2

print(a + b)   # Addition
print(a * b)   # Multiplication
print(a / b)   # Division
```

3. Lists (Arrays)

```
signal = [1, 2, 3, 4, 5]
print(signal[0])   # First element
```

4. NumPy Arrays

```
import numpy as np

x = np.array([1, 2, 3, 4])
print(x * 2)
```

5. Plotting a Signal

```
import numpy as np
import matplotlib.pyplot as plt

t = np.linspace(0, 1, 100)
x = np.sin(2 * np.pi * t)

plt.plot(t, x)
plt.xlabel('Time')
plt.ylabel('Amplitude')
plt.title('Sine Wave')
plt.grid()
plt.show()
```

Lab Tasks

Task 1: Basic Operations

- Write a Python program to:
  - Add two numbers
  - Multiply two numbers
  - Print the results

Task 2: Working with Lists

- Create a list of 5 numbers
- Print:
  - First element
  - Last element
  - Sum of all elements

Task 3: NumPy Practice

- Create a NumPy array of values from 0 to 9
- Multiply the array by 3
- Display the result

Task 4: Generate a Discrete Signal

- Generate a discrete-time signal:

$$x[n] = n, \quad n = 0, 1, 2, \ldots, 10$$

- Print the values

Task 5: Plot a Sine Wave

- Generate a sine wave
- Plot the signal using Matplotlib
- Label axes and title